

Training Configuration and Summary:

Dataset used: rssi_odom_dataset-ORIGINAL.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 16:52:53

Training Start: 2025-05-07 16-15-19

Training End: 2025-05-07 16-40-09

Training Duration: 24.83 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: relu

Optimizer: adam

Dropout Rate: 0.20

MAE: 2.5442

MSE: 9.3711

RMSE: 3.0612

R2 Score: -0.5980

Multi-Step Forecast:

Forecast Steps: 40

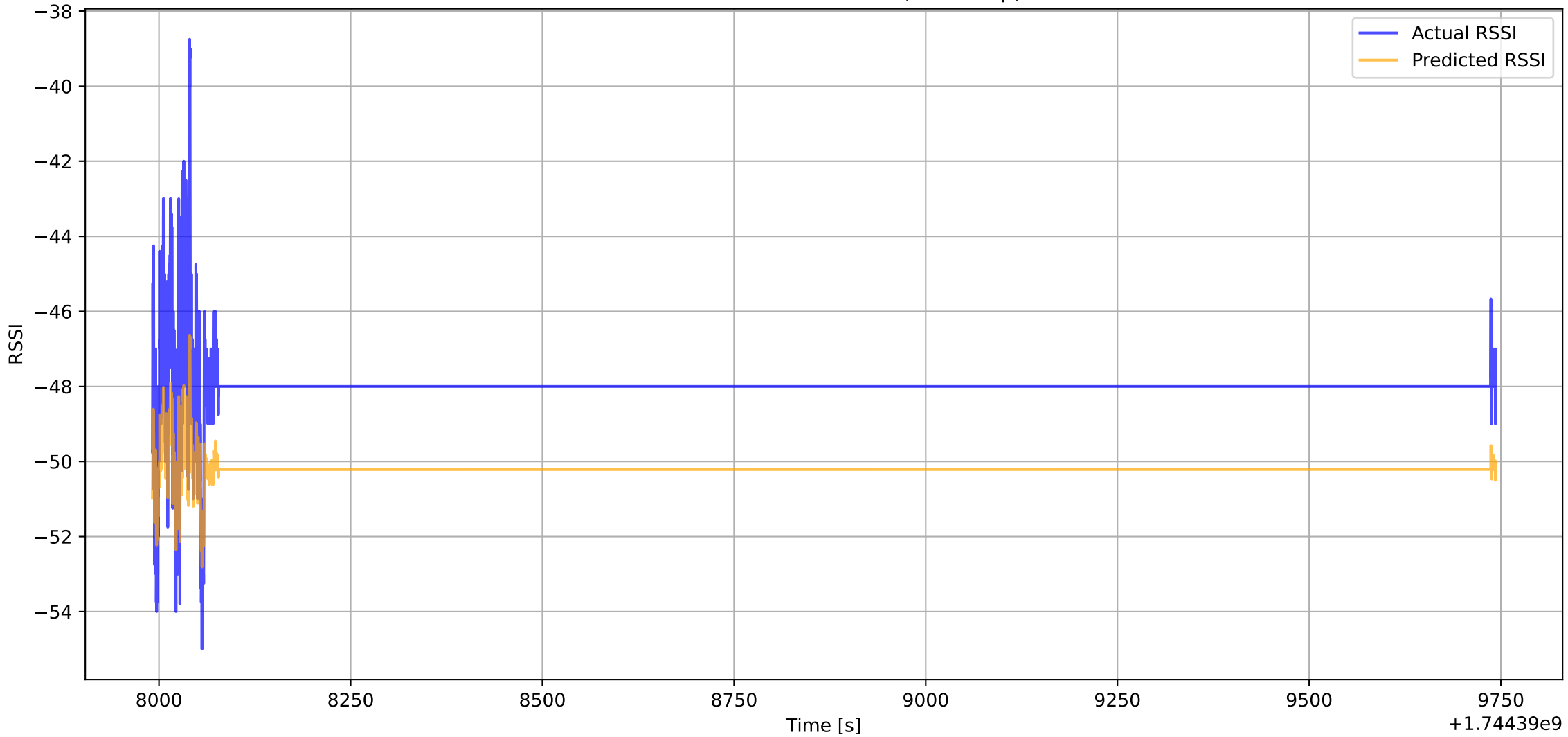
MAE: 0.0365

MSE: 0.0017

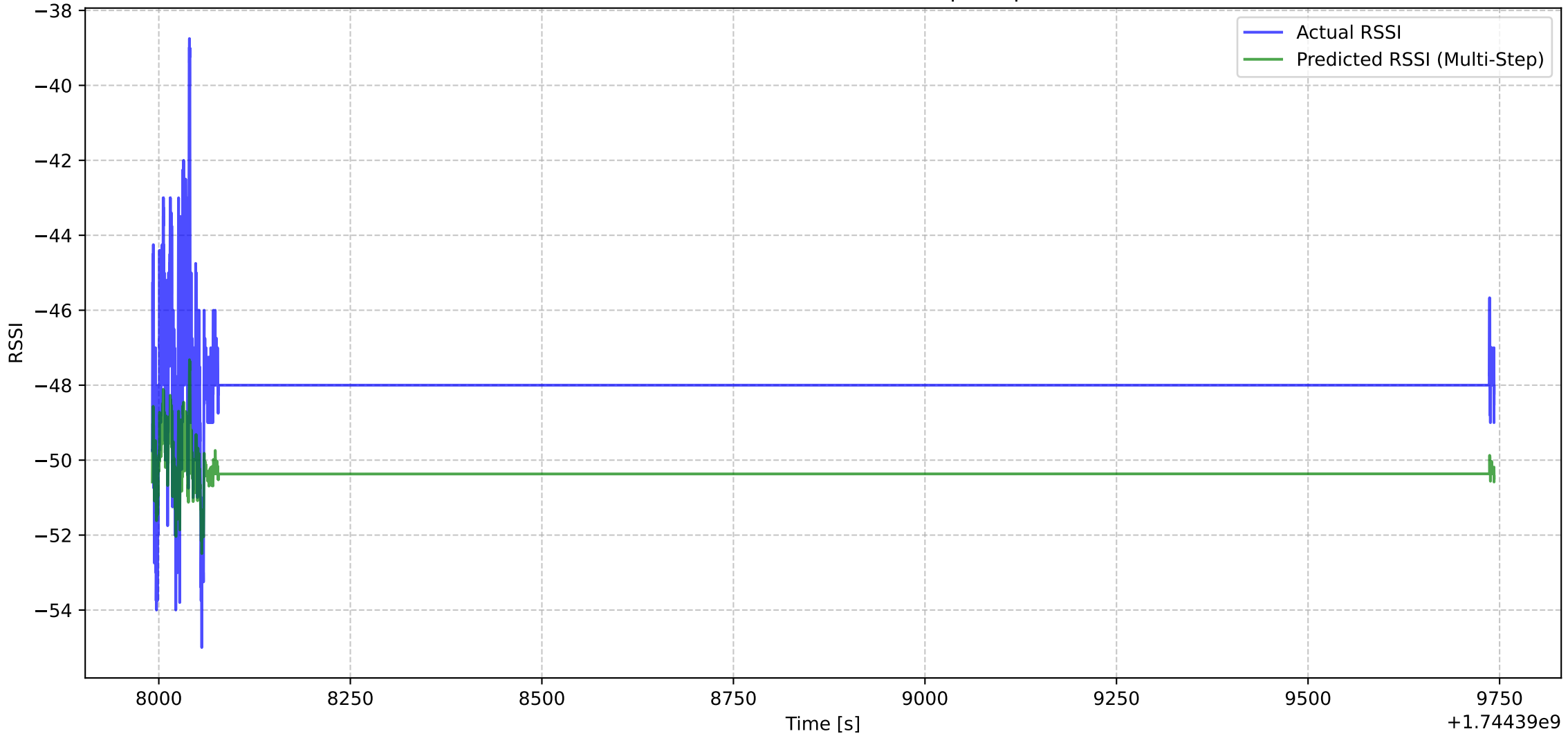
RMSE: 0.0416

R2 Score: -0.2688

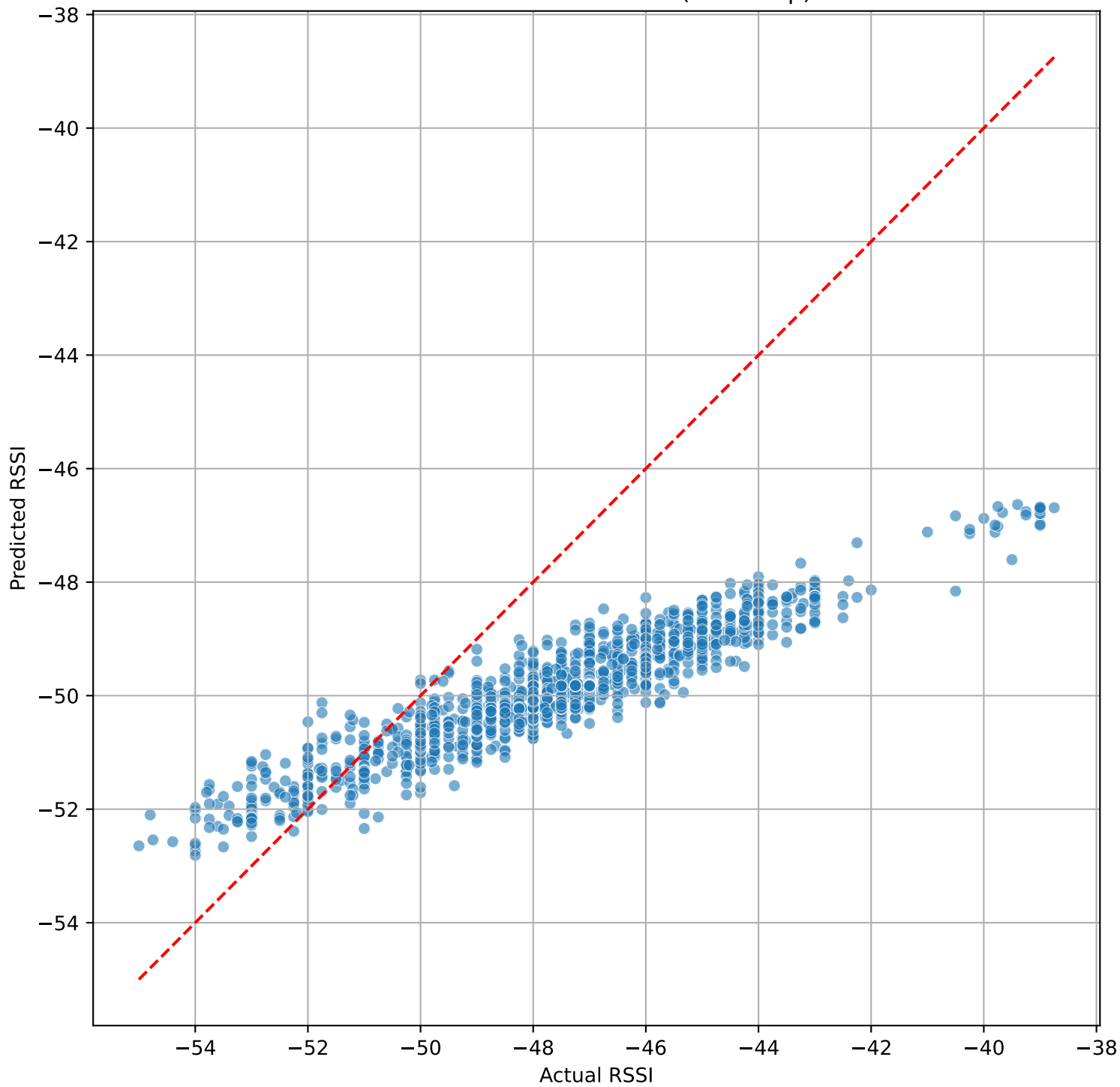
Actual vs Predicted RSSI (First Step)



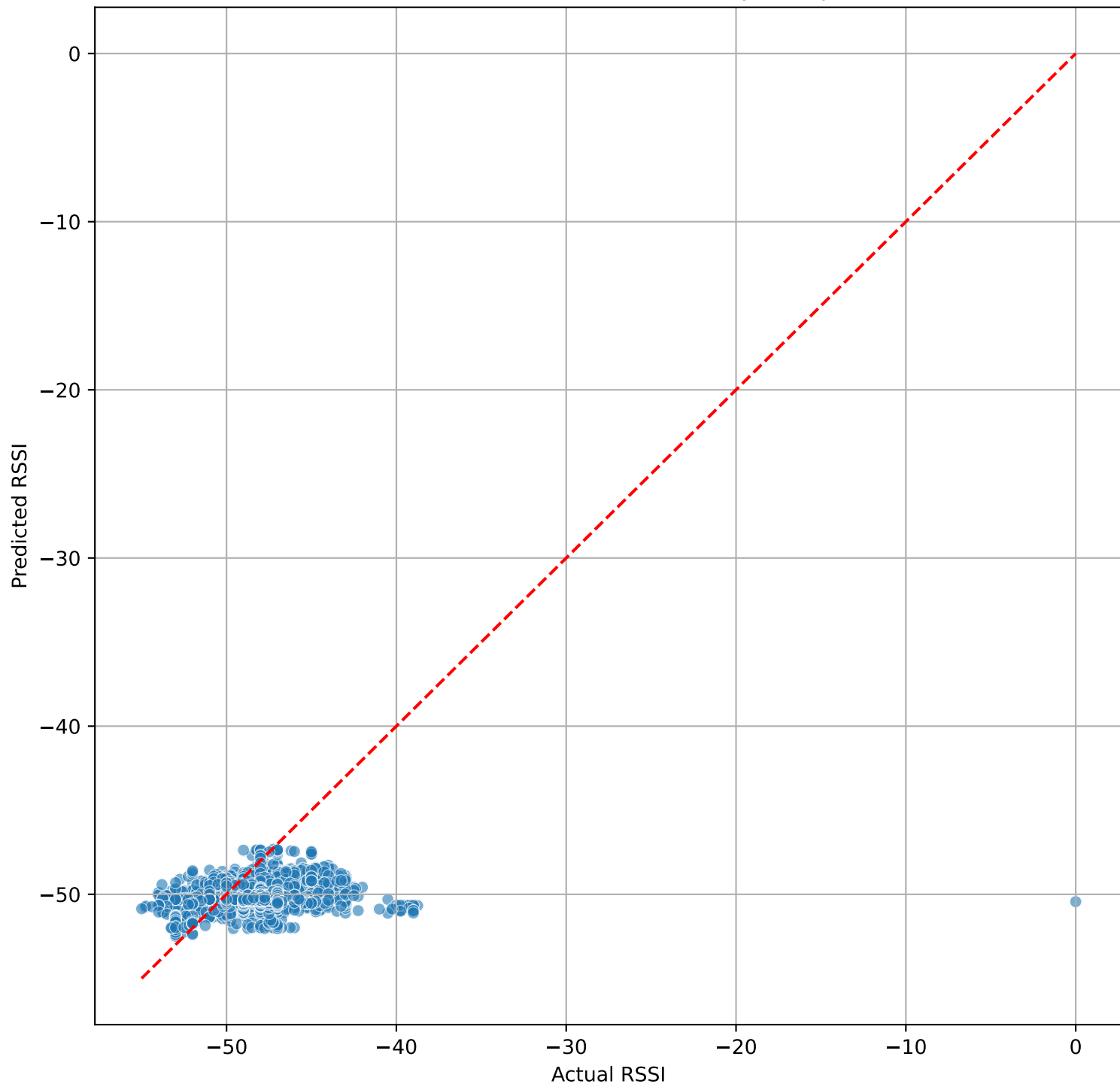
Actual vs Predicted RSSI (Multi-Step (step 40))



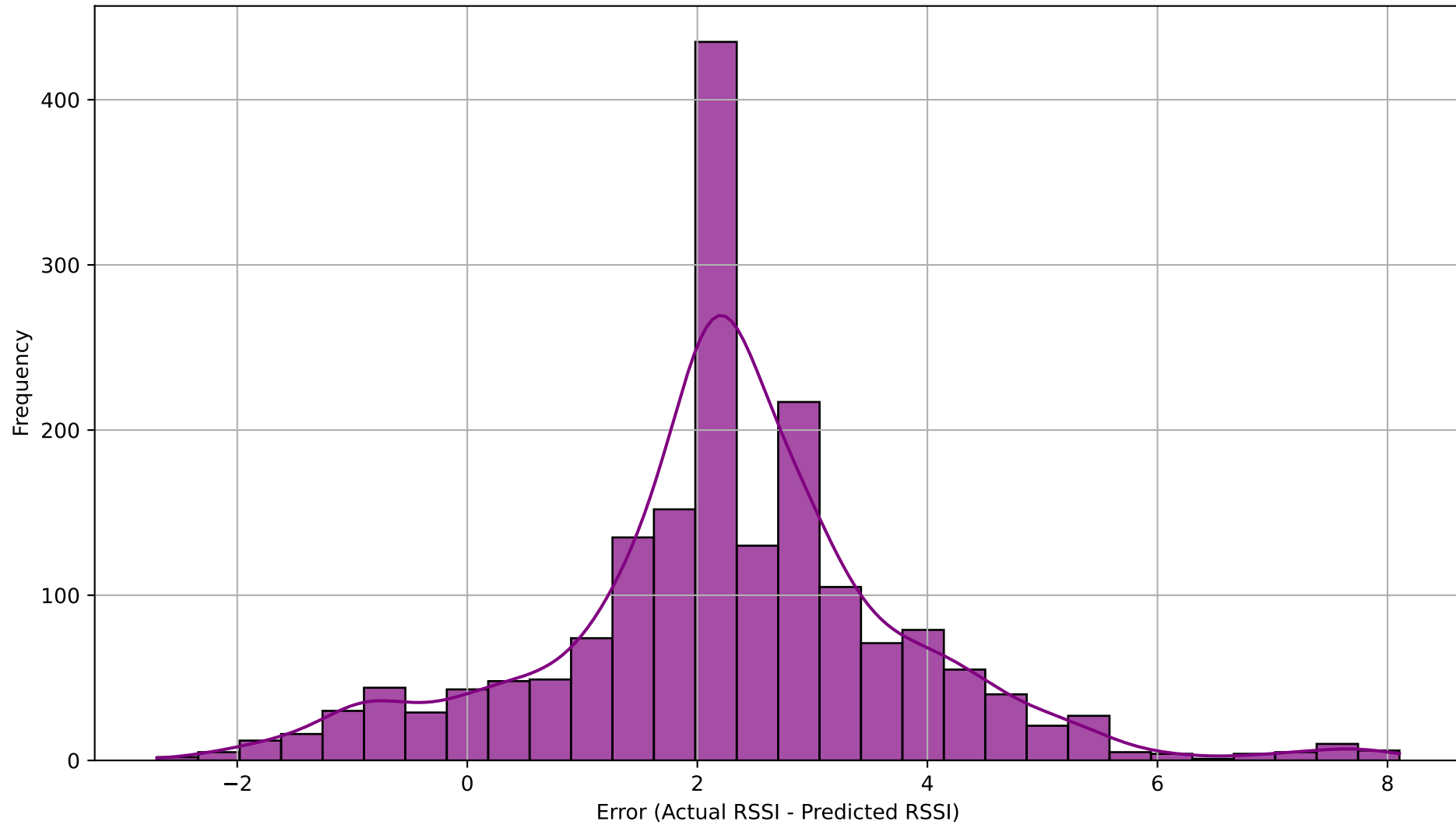
Actual vs Predicted RSSI (First Step)



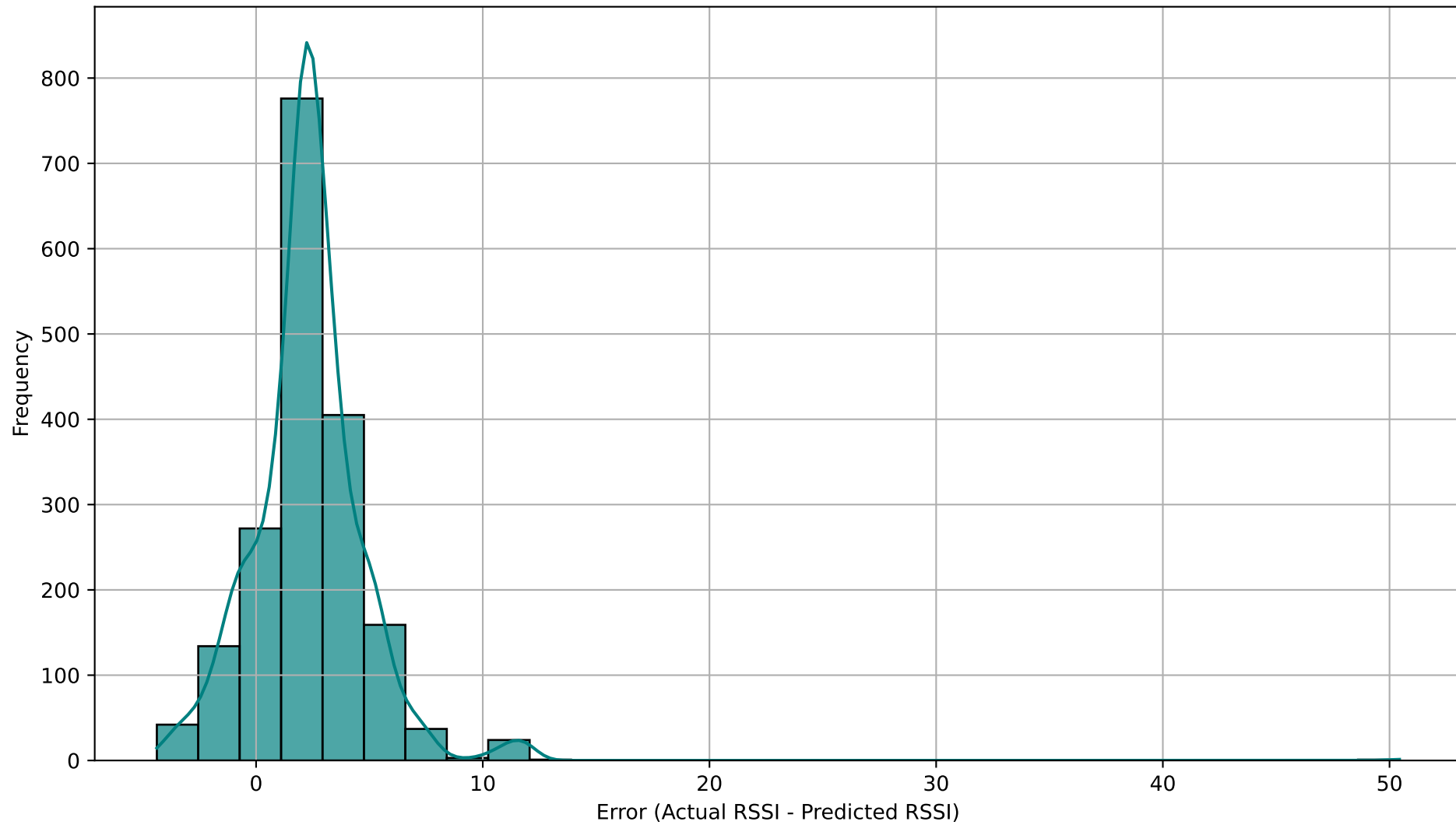
Actual vs Predicted RSSI (Last Step - step 40)



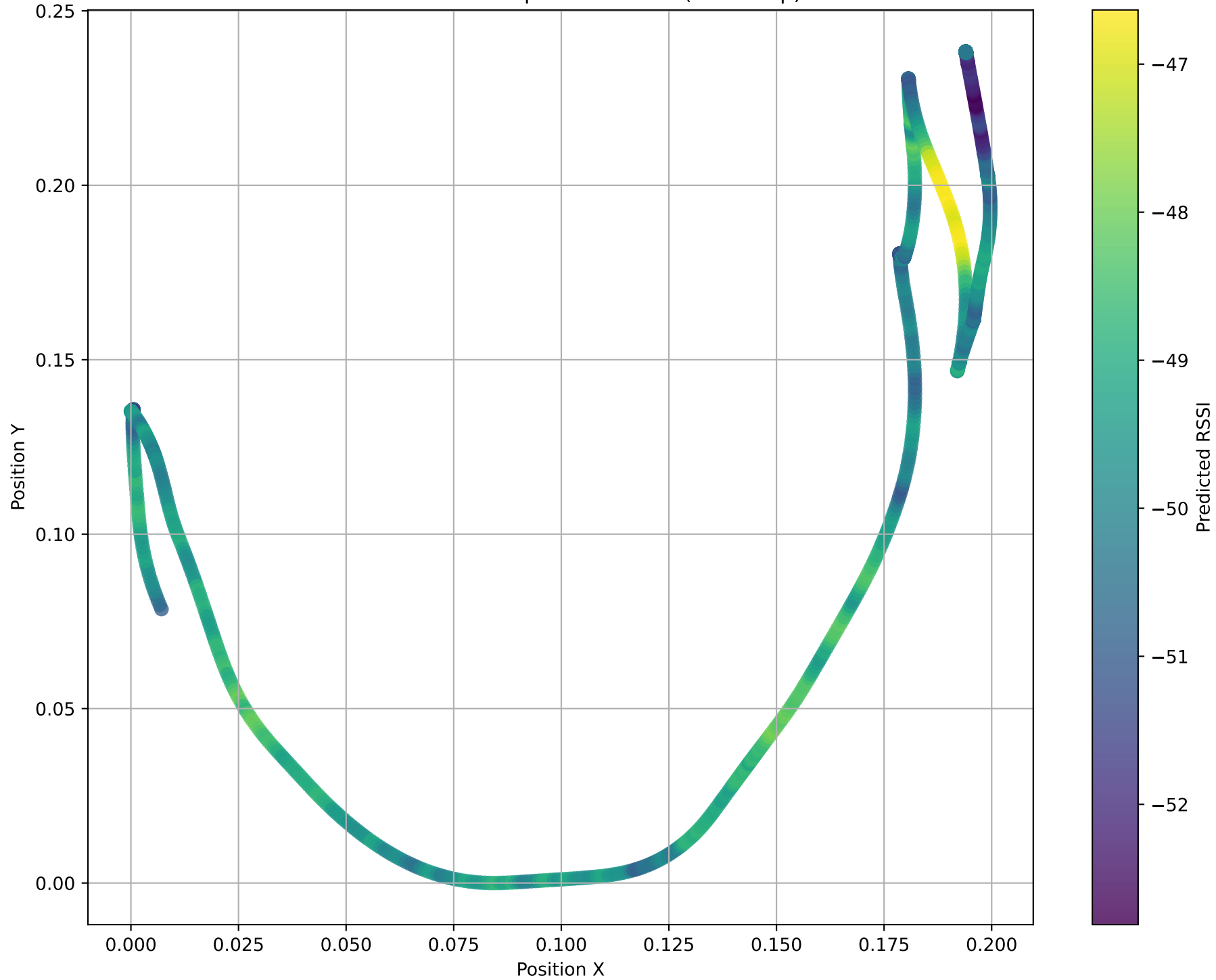
Distribution of Prediction Errors (First Step)



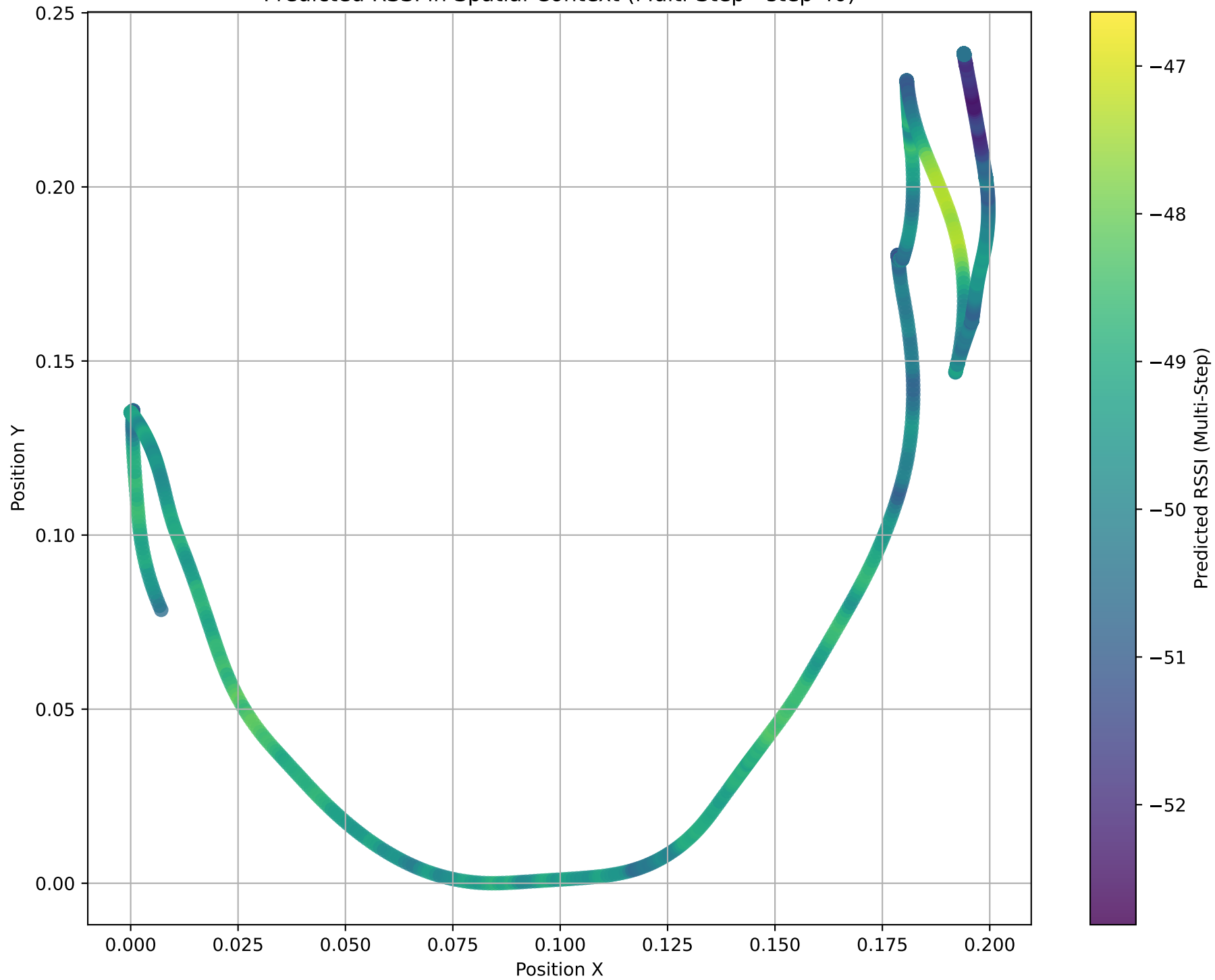
Distribution of Prediction Errors (Last Step - step 40)



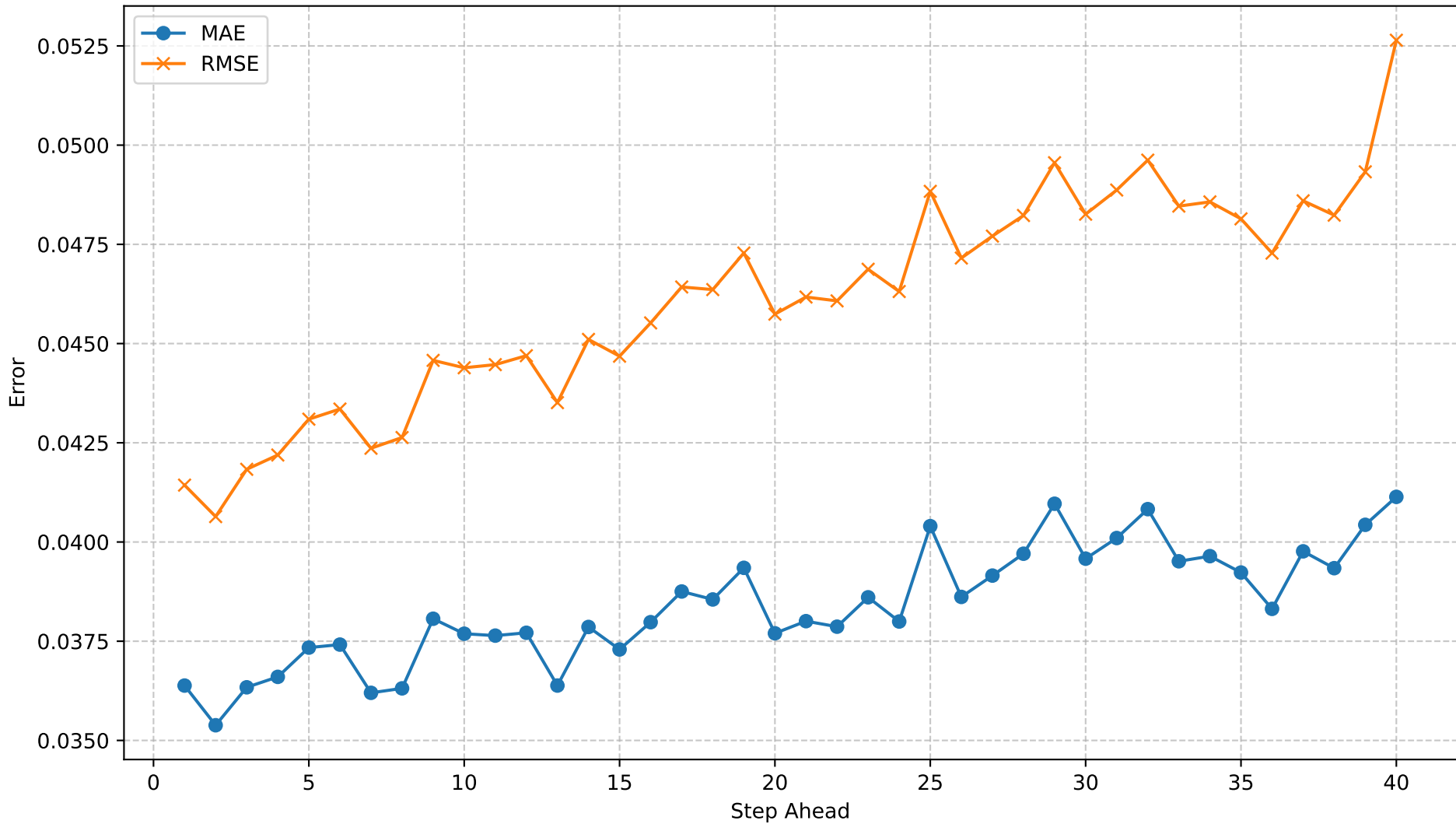
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset-ORIGINAL.parquet

Modelo: LSTM

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.04
- Mediana : 0.04 / 0.05
- 40º passo : 0.04 / 0.05

Crescimento do erro: 13.1% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 40 (0.04), 29 (0.04), 32 (0.04)

Recomendação:

O modelo é adequado para decisões de hand-over até o horizonte máximo configurado.